MySQL Assignment 2 – Querying Data

Create table

```
create database Employee;
use Employee;
create  table departments( department_id   int auto_increment primary key,
    department_name varchar(30) not null);
   create table  location( location_id int  auto_increment primary key,
   location_name varchar(30) not null);
     create table employee( employee_id int auto_increment primary key,
     employee_name varchar(30) not null,
     gender  enum('M','F'),
     age  int  check(age>=18),
     hire_date date default (current_date),
     designation  varchar(50),
     salary decimal(8,2),
     department_id int,
     location_id int,
     foreign key (department_id) references departments(department_id),
     foreign key(location_id) references location (location_id));



     INSERT INTO departments (department_id, department_name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
```

Insert table

- ```
  INSERT INTO departments (department_id, department_name) VALUES
  (1, 'Software Development'),
  (2, 'Marketing'),
  (3, 'Data Science'),
  (4, 'Human Resources'),
  (5, 'Product Management'),
  (6, 'Content Creation'),
  (7, 'Finance'),
  (8, 'Design'),
  (9, 'Research and Development'),
  (10, 'Customer Support'),
  (11, 'Business Development'),
  (12, 'IT'),
  (13, 'Operations');
  ```

- ```
  INSERT INTO location (location) VALUES
  ('Chennai'),
  ('Bangalore'),
  ('Hyderabad'),
  ('Pune');
  ```

- ```
  INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary) VALUES
  (5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
  (5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
  (5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
  (5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
  ```
  ```

```
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),
(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),
(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),
(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),
(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),
(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),
(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),
(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),
(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),
(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),
(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),
(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);
```

Select departments ,employee, location

```
select * from departments;
select * from locations;
select * from employees;


DIATINCT SALARIES FRO EMPLOYEES
select distinct salary from employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Department_id	department_name
1	HR
2	Account
3	IT
4	Field
5	IT
6	HR
7	Account
8	IT
9	Field
10	IT
11	hr
12	account
13	It
14	field

departments 1 × locations 2 employees 3 Apply

```sql
select * from departments;
select * from locations;
select * from employees;


DIATINCT SALARIES FRO EMPLOYEES
select distinct salary from employees;
```

sult Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

location_id	location_name
1	chennai
2	T.Nager
3	Avadi
4	chennai
5	villavakam
NULL	NULL

Alise using age and salary

```sql
select distinct salary from employees;

--- alise using age and salary

select age as employee_age,
salary as employee_salary
from employees;

--- where clause and operator

select  * from employees
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
80000.00
40000.00
60000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

employees 6 ×   Read Only

```
--
--- alise using age amd salary

select age as employee_age,
  salary as employee_salary
  from employees;

--- where clause and operator

select  * from employees
```

employee_age	employee_salary
21	80000.00
24	80000.00
23	40000.00
22	80000.00
21	80000.00
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00

```
select distinct salary from employees;

--- alise using age amd salary

select age as employee_age,
salary as employee_salary
from employees;

--- where clause and operator

select  * from employees
```

salary
80000.00
40000.00
60000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

Display salary

Diisplay salery is greaterthan 5000 and hire date in less than 2016 using where clause

```sql
select  * from employees
 where salary >50000
 and hire_date <'2016-01-01';

    select * from employees where   designation  is null;
   --sorting and grouping
   order by
    select * from employees order by Department_id ASC , salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	Hire_date	designation	salary	Department_id	location_id
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	80000.00	12	1
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	55000.00	11	1
5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	65000.00	8	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Display designation is null

```sql
   select * from employees where   designation  is null;
  --sorting and grouping
  order by
   select * from employees order by Department_id ASC , salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	Hire_date	designation	salary	Department_id	location_id
5004	Kiara Malhotra	F	29	2015-07-05	NULL	70000.00	8	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

display asc and desc using order by

```sql
  order by
   select * from employees order by Department_id ASC , salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	Hire_date	designation	salary	Department_id	location_id
1	ram	M	21	2021-08-07	chennai	80000.00	NULL	NULL
2	sam	M	24	2022-06-02	hyder	80000.00	NULL	NULL
4	RAVI	M	22	2020-08-07	chennai	80000.00	NULL	NULL
5	RAVI	M	21	2021-08-07	chennai	80000.00	NULL	NULL
3	usha	F	23	2021-02-01	chennai	40000.00	NULL	NULL
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	90000.00	2	3
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	60000.00	2	1
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	100000.00	3	4
5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	75000.00	3	2
5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	60000.00	3	4
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	100000.00	4	4
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4

employees 10 ×

Display using limit

```
    limit
    select * from employees where year(Hire_date)= '2018'limit 5;

    Aggregate Functions
    SELECT SUM(salary) AS total_finance_salary
FROM employees
WHERE department_id = (
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

employee_id	employee_name	gender	age	Hire_date	designation	salary	Department_id	location_id
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	55000.00	4	4
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	90000.00	1	1
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	85000.00	7	3
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	60000.00	4	4
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	70000.00	2	2
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

employees 11 ×

```
SELECT MIN(age) AS minimum_age
FROM employees;

group by
list of max
select location_id ,max(salary) max_salary
from employees
group by  location_id;

calculate  average
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

minimum_age
21

Result 13 ×

```
 select distinct salary from employees;

 --- alise using age amd salary

 select age as employee_age,
 salary as employee_salary
 from employees;

--- where clause and operator

select  * from employees
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

salary
80000.00
40000.00
60000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

employees 6 ×   Read Only

Display groupby unction using max salray

```
group by
list of max
select location_id ,max(salary) max_salary
from employees
group by  location_id;

calculate  average
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	max_salary
NULL	80000.00
1	95000.00
2	95000.00
3	90000.00
4	100000.00

Result 14 | Read Only

Display where clause and group by to find designation by average salary

```
select designation,
avg (salary) as average_salary
from employees
where designation  like '% Analyst%'
group by designation;

--Having

SELECT department_id, COUNT(*) AS employee_count
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

designation	average_salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

Result 15

30°C Partly cloudy

Count all employee group by departments using id

```sql
SELECT department_id, COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
HAVING COUNT(*) < 3;


SELECT location_id, AVG(age) AS average_age
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	employee_count
1	1
5	1
6	2
7	2
9	1
11	2
13	1

Result 16 ×

Display gender wise feamale averqage age

```sql
SELECT location_id, AVG(age) AS average_age
FROM employees
WHERE gender = 'F'
GROUP BY location_id
HAVING AVG(age) < 30;

--join functions using
inner join
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

location_id	average_age
NULL	23.0000
1	26.7500
2	27.3333
4	29.2000

esult 17 × Read

Left join

```sql
SELECT
    l.location_name,
    e.employee_name
FROM locations AS l
LEFT JOIN employees AS e
    ON l.location_id = e.location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_name	employee_name
chennai	Reyansh Singh
chennai	Anvi Chaudhary
chennai	Saanvi Patel
chennai	Ishaan Kumar
chennai	Aarav Nair
chennai	Aryan Shetty
chennai	Anaya Kapoor
chennai	Sara Iyer
T.Nager	Aaradhya Iyer
T.Nager	Dhruv Shetty
T.Nager	Myra Verma
T.Nager	Vihaan Mohan
T.Nager	Kabir Nair
T.Nager	Anika Paul

Result 26

Inner join

```sql
select e.employee_name,
e.designation,
d.Department_name
from employees as e
inner join departments AS d
  on e.department_id= d.department_id;

Lefgt join
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	Department_name
Vihaan Singh	Data Analyst	IT
Reyansh Singh	Network Engineer	account
Aaradhya Iyer	Customer Support Executive	IT
Kiara Malhotra	NULL	IT
Anvi Chaudhary	Business Development Executive	hr
Dhruv Shetty	UI Developer	IT
Anushka Singh	Marketing Manager	Account
Diya Jha	Graphic Designer	IT
Kiaan Desai	Sales Executive	hr
Atharv Yadav	Systems Administrator	account
Saanvi Patel	Marketing Analyst	Account
Myra Verma	Operations Manager	It
Arnav Rao	Customer Success Manager	IT
Vihaan Mohan	Supply Chain Analyst	IT

Result 18 Read Only